Established – 1961 Subject: __________

# SEVA SADAN'S

# R. K. TALREJA COLLEGE

# OF

# ARTS, SCIENCE & COMMERCE

# ULHASNAGAR – 421 003

## CERTIFICATE

**This is to certify that Mr./Ms. ______________________________ of S.Y. Computer Science (SYCS) Roll No. ____________________________ has satisfactorily completed The Internet Of Thing Mini Project entitled________________ ________________________________________ during the academic year 2025 – 2026, as a part of the practical requirement. The project work is found to be satisfactory and is approved for submission.**

**PROF. INCHARGE** **HEAD OF DEPT**

______________ ______________

# INDEX

# INTRODUCTION

RFID Based Attendence System- Introduction: In schools, colleges, and organizations, attendance is usually recorded manually using registers or sign sheets. This traditional method is time-consuming, prone to errors, and allows unfair practices such as proxy attendance. Managing and maintaining attendance records also becomes difficult when the number of students or employees is large. Therefore, there is a need for an automated, accurate, and efficient attendance system. Purpose of the Project The purpose of the Smart Attendance System using RFID is to automate the attendance process by using Radio Frequency Identification (RFID) technology. Each person is provided with an RFID card, and attendance is marked automatically when the card is scanned. The system reduces human effort, saves time, and ensures reliable data storage using IoT technology. Importance and Real-Life Applications This system plays an important role in improving efficiency and accuracy in attendance management. It eliminates manual work and reduces chances of fraud.

- Motivation: Traditional attendance systems in schools, colleges, and organizations are often manual, time-consuming, and prone to errors. Manual attendance marking can lead to issues such as proxy attendance, inaccurate record-keeping, data loss, and unnecessary administrative workload.
  With advancements in automation and wireless identification technologies, systems like Radio Frequency Identification (RFID) provide an efficient and reliable alternative. An RFID-based attendance system enables automatic identification and recording of attendance with minimal human intervention.

- Problem Definition : Traditional attendance systems used in schools, colleges, and organizations rely on manual methods such as paper registers or spreadsheets, which are time-consuming, prone to human error, and vulnerable to issues like proxy attendance and data mismanagement. Maintaining and retrieving records manually also increases administrative workload and makes real-time monitoring difficult. Therefore, there is a need for an automated, accurate, and secure solution that minimizes manual intervention and improves efficiency. By using Radio Frequency Identification (RFID) technology, attendance can be recorded automatically when authorized cards are scanned, ensuring reliable data storage, reducing fraud, and streamlining the overall attendance management process.

**How it works**: he RFID-based attendance system works using Radio Frequency Identification (RFID) technology to automatically record attendance. Each student or employee is provided with a unique RFID card containing an embedded chip with a specific identification number. When the card is brought near the RFID reader, the reader scans the card and sends the unique ID to a microcontroller (such as Arduino or another controller used in the project). The system then verifies the ID with the stored database, and if it matches, the attendance is recorded with the date and time. The recorded data is stored in a database or displayed on a screen, ensuring accurate, fast, and automated attendance tracking without manual intervention.

- **Key Features :** Automated Attendance Recording – Attendance is marked automatically when an RFID card is scanned, eliminating manual entry.
- Fast and Contactless Operation – Uses Radio Frequency Identification (RFID) technology for quick and touch-free identification.
- Unique User Identification – Each user is assigned a unique RFID tag to ensure accurate tracking.
- Prevention of Proxy Attendance – Reduces chances of fake or duplicate attendance marking.
- Secure Data Management – Records can be securely stored in a database for easy retrieval.
- Reduced Administrative Workload – Minimizes paperwork and manual calculations.
- Scalable System – Can be implemented in schools, colleges, offices, and other organizations.

**Scope of the Project :** The scope of the RFID-Based Attendance System is to design and implement an automated solution for recording and managing attendance using Radio Frequency Identification (RFID) technology. The system is intended for use in educational institutions, offices, and organizations to replace traditional manual attendance methods with a faster, more accurate, and secure process. It includes RFID card identification, real-time attendance recording, and secure data storage for easy monitoring and report generation. The project can be further expanded by integrating features such as cloud storage, SMS/email notifications, biometric verification, and web-based monitoring systems, making it scalable and adaptable for future enhancements.

**Objectives:**

- To design and implement an automated attendance management system using Radio Frequency Identification (RFID) technology for accurate user identification.
- To develop a system that assigns a unique RFID tag to each user for secure and reliable attendance tracking.
- To interface the RFID reader with a microcontroller (such as Arduino/NodeMCU) for real-time data processing.
- To record attendance automatically along with date and time stamps to ensure precise tracking.
- To create a structured database for storing, managing, and retrieving attendance records efficiently.
- To reduce human errors, paperwork, and administrative workload associated with manual attendance systems.
- To prevent proxy attendance by validating unique RFID card IDs.
- To design a system that can be expanded with additional features such as cloud storage, web interface, biometric integration, or notification systems.
- To ensure the system is cost-effective, user-friendly, and scalable for implementation in schools, colleges, and organizations.

# REQUIREMENT SPECIFICATION

Hardware Components:

| Component | Specification |
|---|---|
| Microcontroller | Arduino Uno |
| RFID Module | Used to read card |
| RFID Card | Holds an unique UID number |
| LCD Display | To Display The Output |
| I2C Module | Reduce wiring Connections |
| Buzzer | For Sounding |

Software Requirements:

Arduino Uno IDE – Software to programming the adruino board

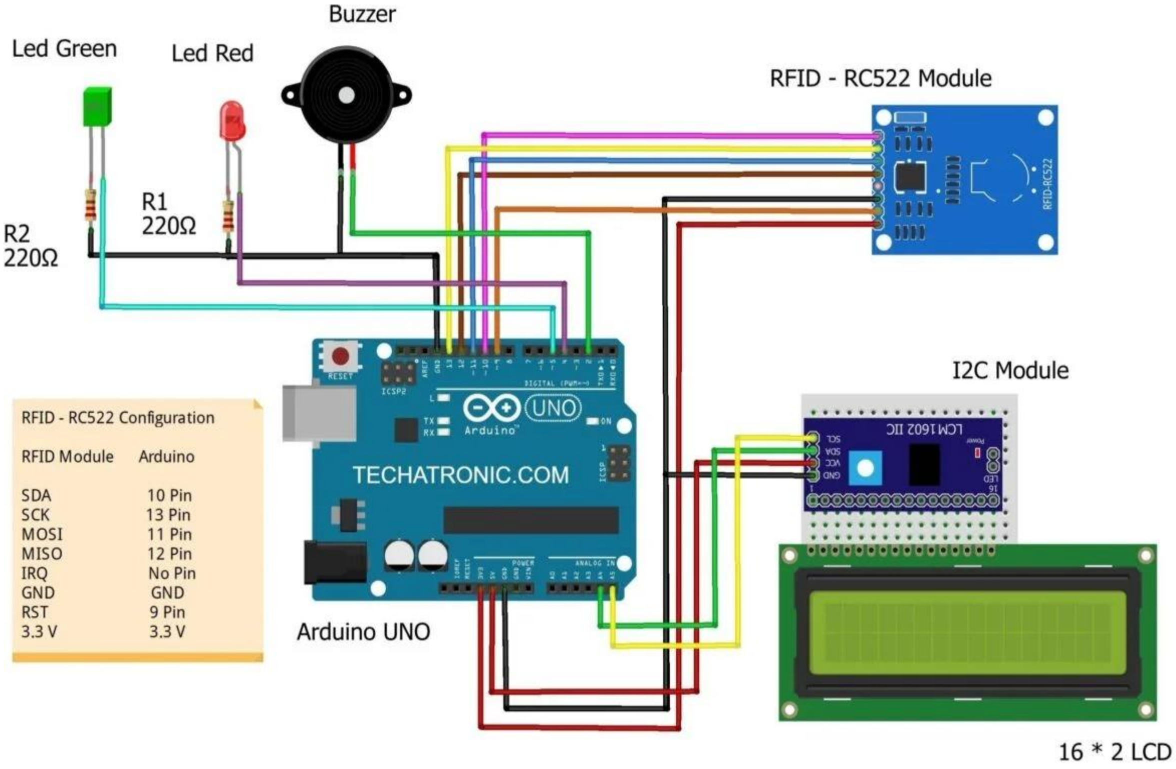
# SYSTEM DESIGN

Block Diagram:

```
+--------------------+
        |    RFID Card/Tag   |
        +--------------------+
                 ↓
        +--------------------+
        |   RC522 RFID       |
        |     Reader         |
        +--------------------+
                 ↓ (SPI Communication)
        +--------------------+
        |     Arduino UNO    |
        |  (Main Controller) |
        +--------------------+
          ↓       ↓       ↓
    +-----------+  +------+  +---------+
    | 16x2 LCD  |  |Buzzer|  |  LEDs   |
    | (I2C)     |  |      |  |Red/Green|
    +-----------+  +------+  +---------+
          ↓
    +-----------------+
    | Attendance Count |
    |  & UID Database  |
    +-----------------+
```

- RFID Card is tapped.
- RC522 Reader reads the UID.
- UID is sent to Arduino UNO.
- Arduino:
- Checks UID with stored IDs.
- Marks attendance.
- Increases total count.
- Shows name + total on LCD.

Circuit Diagram:




Description: This circuit is designed to scan RFID cards and mark attendance.

- How It Works:

The RC522 RFID module reads the UID of the RFID card.
Arduino UNO processes the UID.
If UID matches:
Green LED turns ON
Buzzer beeps
LCD displays “Welcome [Name]”
If UID does not match:
Red LED turns ON
Buzzer gives different tone
LCD displays “Access Denied”

Key Steps of Circuit Diagram:

Step 1: Power Supply
Arduino provides:
5V for LCD
3.3V for RFID
Common GND for all components

Step 2: Initialization
In code:
SPI communication starts
RFID module initialized
LCD initialized
Pins set as OUTPUT
Step 3: Card Detection
RFID continuously checks for a card
Reads UID
Step 4: Verification
Arduino compares UID with stored UIDs
Step 5: Output Response
Valid → Green LED + Buzzer + Welcome message
Invalid → Red LED + Different beep + Denied message

Benefits of circuit diagram :

✔ Clear visualization of connections
✔ Prevents wrong wiring
✔ Easy troubleshooting
✔ Helps beginners understand SPI & I2C communication
✔ Can be expanded (add SD card, WiFi, etc.)
✔ Cost-effective attendance solution

When to Use a Circuit Diagram ?

Use a circuit diagram when:
🔧 Building hardware projects
🛠 Troubleshooting wiring issues
📚 Teaching electronics
📄 Submitting project documentation
🧪 Before assembling on breadboard
🏫 Creating academic projects
Without a diagram → High chance of wiring mistakes.

Circuit Model Testing Methods :

1️⃣ Visual Inspection
Check loose wires
Check correct pin numbers
Confirm 3.3V for RFID
2️⃣ Continuity Testing (Using Multimeter)
Check if GND is common
Verify no short circuit
3️⃣ Power Testing

Measure voltage:
RC522 → 3.3V
LCD → 5V

4️⃣ Component-wise Testing

Test individually:

✔ Upload LED blink code
✔ Test buzzer separately
✔ Run I2C scanner for LCD
✔ Run RFID dump code

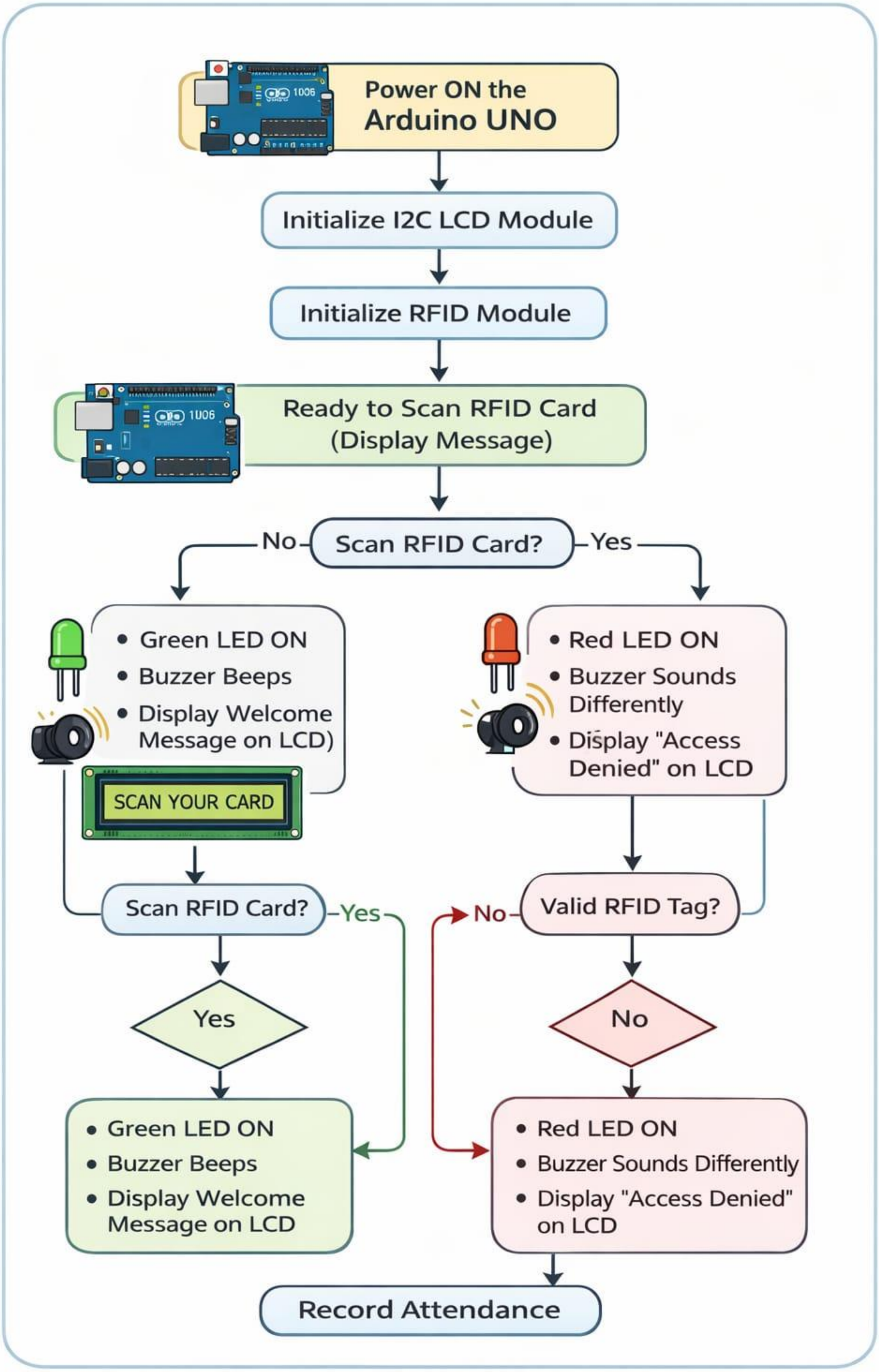
5️⃣ Software Debug Testing

Use Serial Monitor:
Print UID
Check matching logic
Verify attendance counter

6️⃣ Stress Testing

Scan multiple cards repeatedly
Check duplicate entry handling
Verify total attendance display

Activity Diagram :



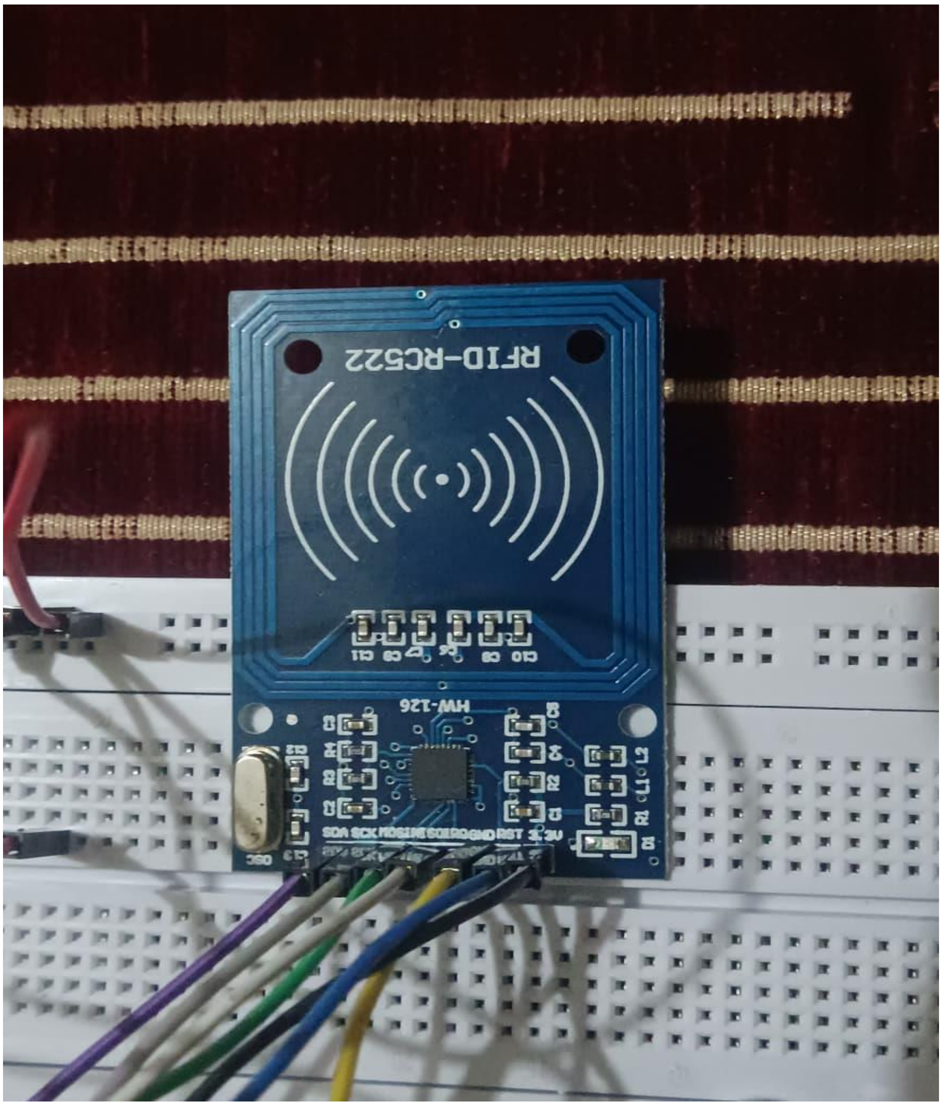
**Description :** The activity diagram represents the working process of an RFID-based Attendance System using Arduino UNO, RC522 RFID module, I2C LCD display, LEDs, and a buzzer. The system starts by powering on the Arduino and initializing the LCD and RFID modules. It then displays a message indicating that it is ready to scan an RFID card. When a card is scanned, the system checks whether the RFID tag is valid. If the tag is valid, the green LED turns on, the buzzer beeps, a welcome message is displayed on the LCD, and attendance is recorded. If the tag is invalid, the red LED turns on, the buzzer sounds differently, and an "Access Denied" message is displayed. The diagram clearly shows the sequence of actions, decision points, and outputs of the system.

## Purpose of Activity Diagrams :

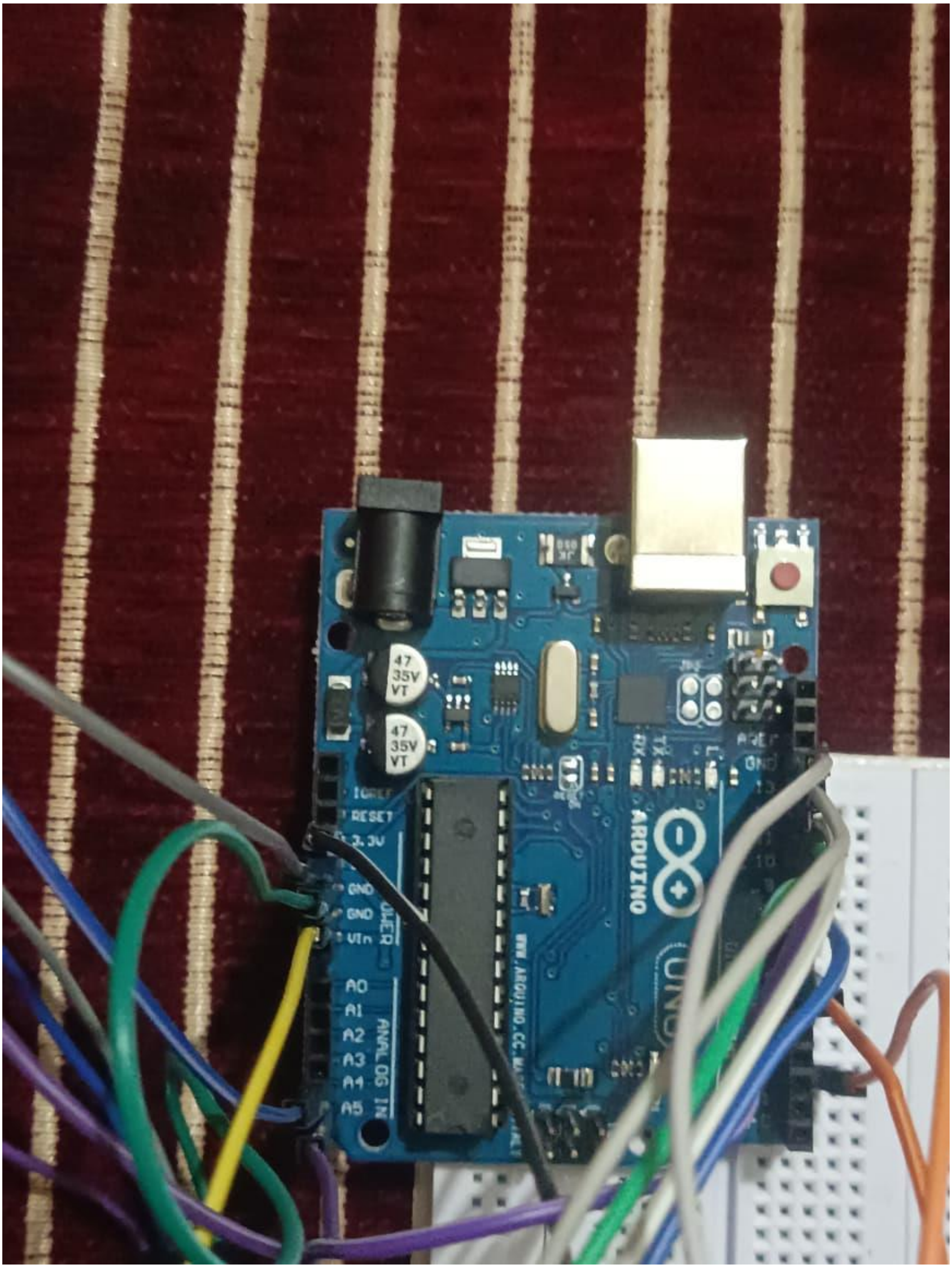
1. To visually represent the workflow of the system.
2. To show the sequence of operations from start to end.
3. To identify decision points (like valid or invalid RFID tag).
4. To explain system logic clearly and simply.
5. To help in understanding program flow before coding.
6. To assist in project documentation and presentation.
7. To detect logical errors in system design.
8. To improve communication between team members.
9. To simplify troubleshooting and debugging.
10. To provide a structured model of system behavior.

# SYSTEM IMPLEMENTATION

Step By Step assemble:

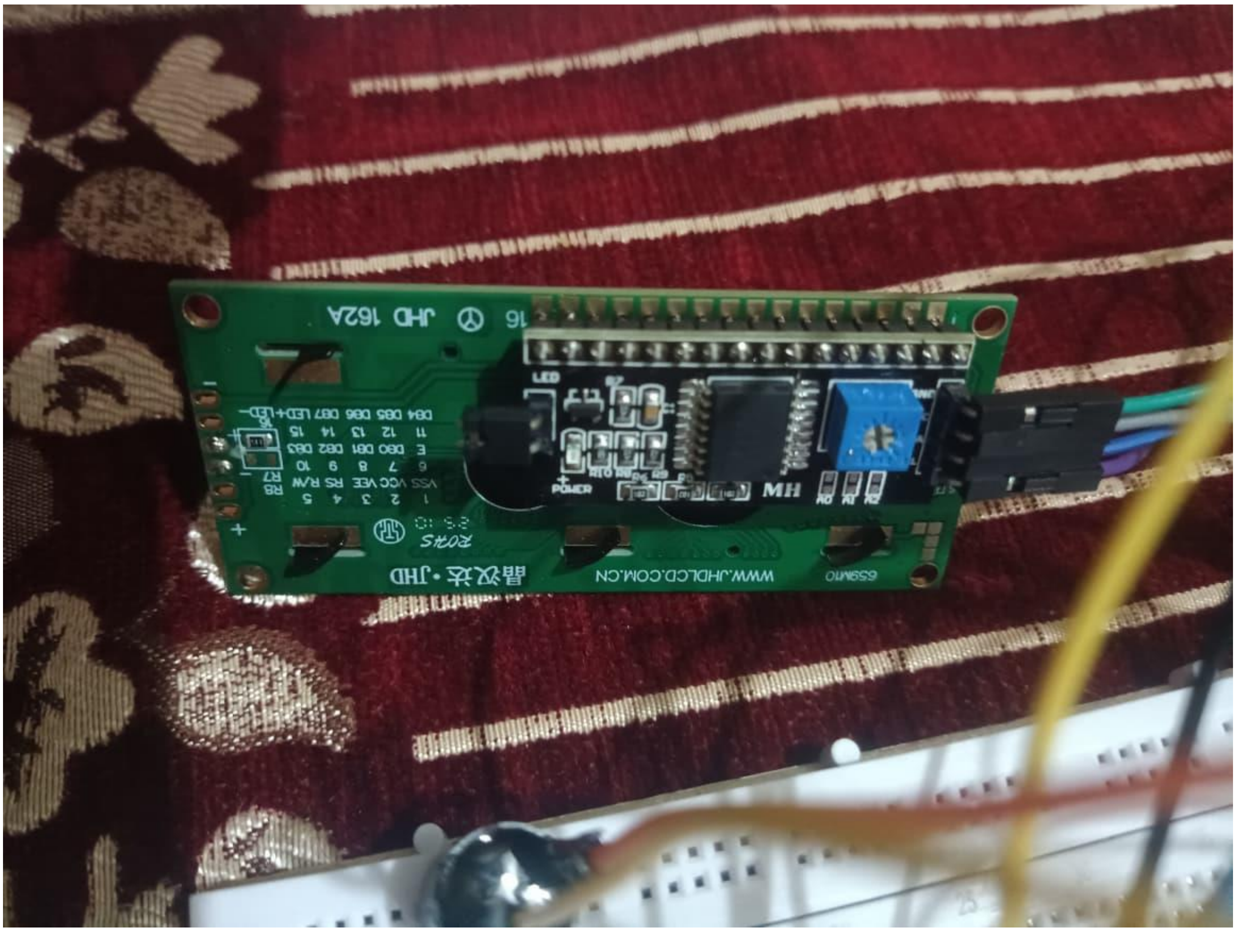



Step 1: Joining Male To Male Jumper wire in the RC522 Module

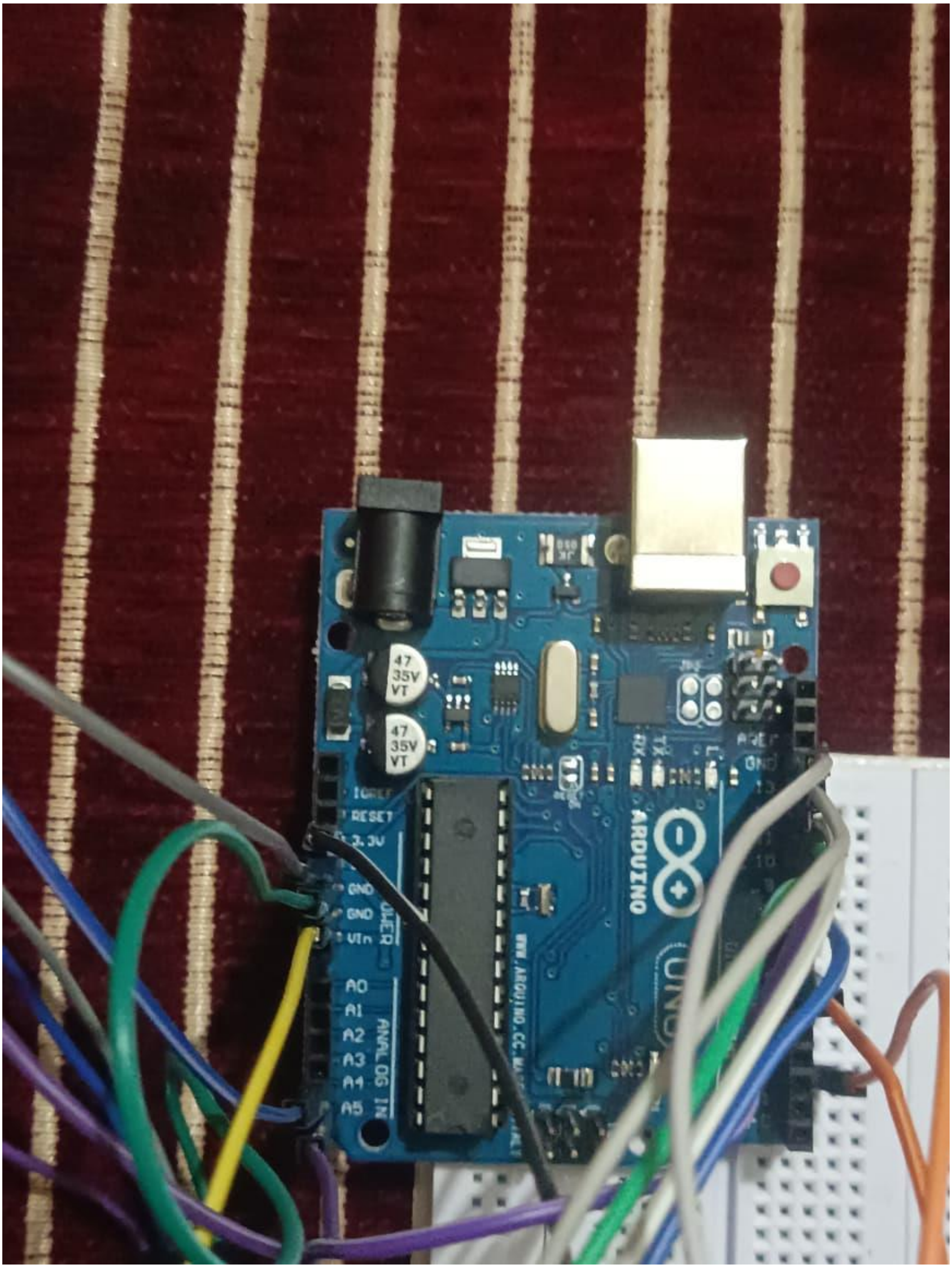



Step 2: Now joining the same wire which we had joined in the RC522 module the other end will be joined in Arduino UNO. As Written Below:

| RFID Module | Arduino UNO |
|---|---|
| SDA | 10 pin |
| SCK | 13 pin |
| MOSI | 11 pin |
| MISO | 12 pin |
| IRQ | No Pin |
| GND | GND |
| RST | 9 pin |
| 3.3V | 3.3 V |

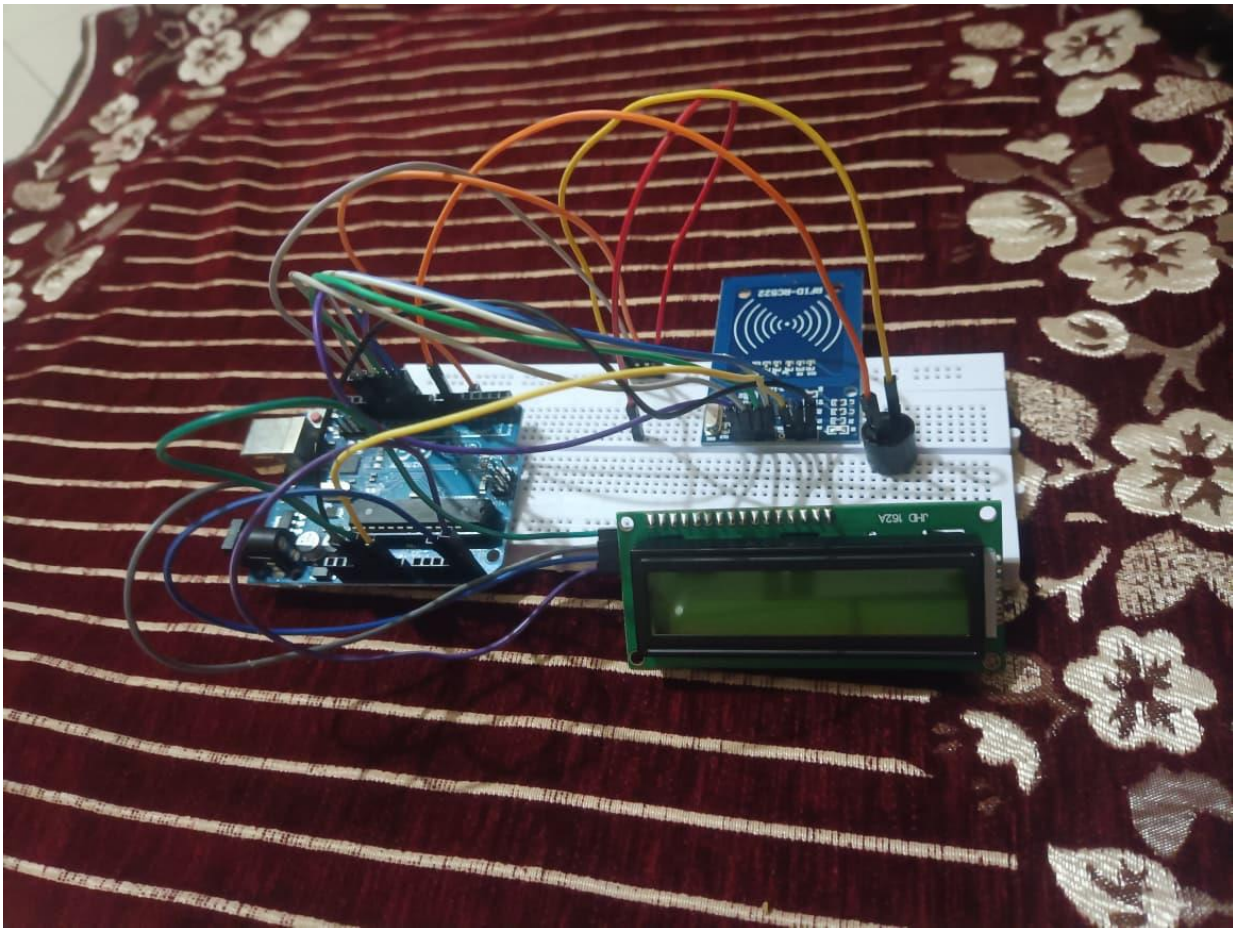


Step 3: Now We will join Male to Female Jumper Wires in I2C Module which is attached in LCD Display.




Step 4: Again will Join the I2C module wire in Arduino UNO accordingly. As written Below:

| I2C Module | Arduino UNO |
|---|---|
| GND | GND |
| VCC | 5 v |
| SDA | A-4 |
| SCL | A-5 |



Step 5: Completed!

Step 6: Now Lets Create A Programme so that is can perform.

Code:

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define SS_PIN 10
#define RST_PIN 9
#define BUZZER 2

MFRC522 mfrc522(SS_PIN, RST_PIN);
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
String uidSunny = "97 13 B4 51";
String uidRahul = "11 22 33 44";
String uidAmit  = "62 63 BA 5C";

// ====== Attendance Status ======
bool sunnyMarked = false;
bool rahulMarked = false;
bool amitMarked  = false;

int totalPresent = 0;

void setup() {
  Serial.begin(9600);
  SPI.begin();
  mfrc522.PCD_Init();
  lcd.init();
  lcd.backlight();
  pinMode(BUZZER, OUTPUT);

  showMainScreen();
}

void loop() {

  if (!mfrc522.PICC_IsNewCardPresent()) return;
  if (!mfrc522.PICC_ReadCardSerial()) return;

  String content = "";

  for (byte i = 0; i < mfrc522.uid.size; i++) {
    content += String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
    content += String(mfrc522.uid.uidByte[i], HEX);
  }

  content.toUpperCase();
  content.trim();

  lcd.clear();

  // ====== SUNNY ======
  if (content == uidSunny) {
    if (!sunnyMarked) {
      sunnyMarked = true;
      totalPresent++;
      lcd.setCursor(0,0);
      lcd.print("Welcome Sunny");
      beep();
```

```
  } else {
    lcd.setCursor(0,0);
    lcd.print("Already Marked");
  }
}

// ====== RAHUL ======
else if (content == uidRahul) {
  if (!rahulMarked) {
    rahulMarked = true;
    totalPresent++;
    lcd.setCursor(0,0);
    lcd.print("Welcome Rahul");
    beep();
  } else {
    lcd.setCursor(0,0);
    lcd.print("Already Marked");
  }
}

// ====== AMIT ======
else if (content == uidAmit) {
  if (!amitMarked) {
    amitMarked = true;
    totalPresent++;
    lcd.setCursor(0,0);
    lcd.print("Welcome Amit");
    beep();
  } else {
    lcd.setCursor(0,0);
    lcd.print("Already Marked");
  }
}

// ====== UNKNOWN CARD ======
else {
  lcd.setCursor(0,0);
  lcd.print("Access Denied");
}

// Always show total on second line
lcd.setCursor(0,1);
lcd.print("Present: ");
lcd.print(totalPresent);

delay(2000);
showMainScreen();
```

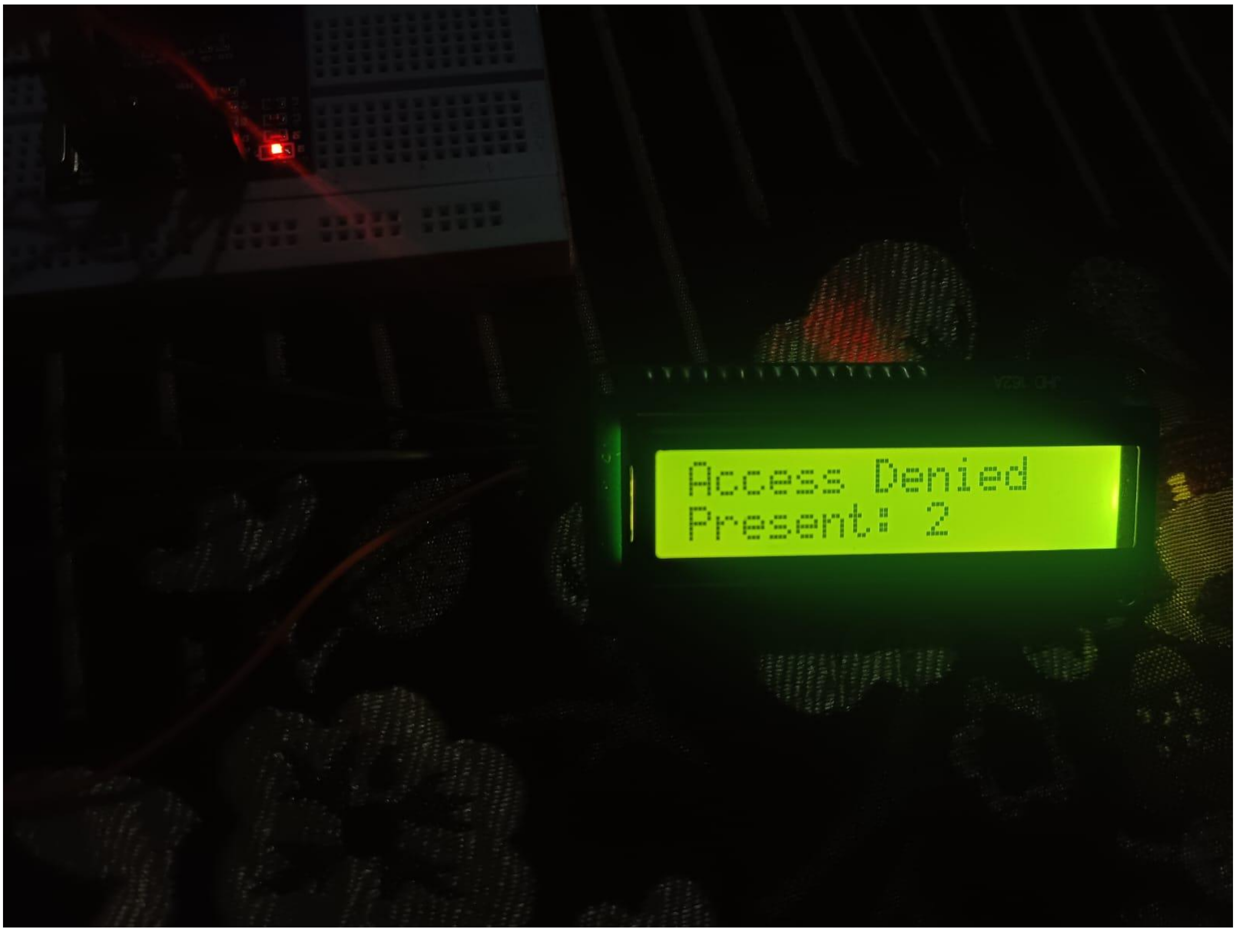
```
  mfrc522.PICC_HaltA();
}

void showMainScreen() {
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Scan Your Card");
  lcd.setCursor(0,1);
  lcd.print("Present: ");
  lcd.print(totalPresent);
}

void beep() {
  digitalWrite(BUZZER, HIGH);
  delay(200);
  digitalWrite(BUZZER, LOW);
}
```

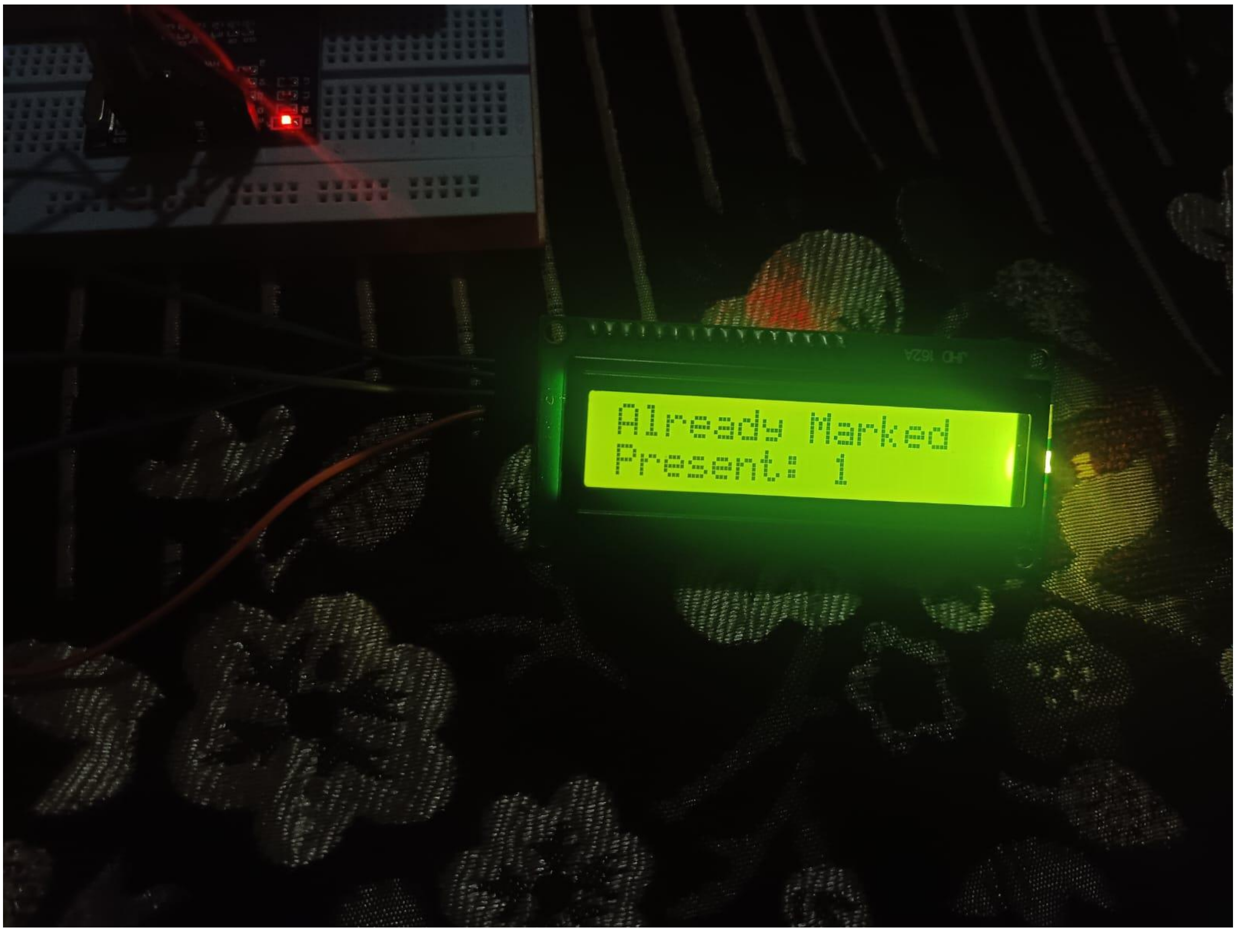
OUTCOME:

Case 1: When a Card whose UID is Not Registered is scanned:



Case 2: when a card with legit UID is Scanned:


Case 3: When a card which I already got scanned try to get scanned again:


Already Marked
Present: 1

# FUTURE SCOPE AND CONCLUSION

**Future Scope :**

- ntegration with **cloud-based storage** for real-time access and secure backup of attendance data.
- Addition of **biometric features** like fingerprint or facial recognition to prevent proxy attendance.
- Implementation of **automated notifications** via SMS or email for students, employees, or administrators.
- Expansion for **large-scale deployment** in universities, corporations, or multi-branch organizations.
- Integration with **payroll or academic performance systems** for complete automation.
- Development of a **web or mobile interface** for easy monitoring and report generation.
- Potential use of **IoT and AI analytics** to track attendance patterns and improve organizational efficiency.

## Conclusion:

The RFID-based attendance system provides an efficient, accurate, and automated solution for managing attendance in schools, colleges, and organizations. By using Radio Frequency Identification (RFID), it eliminates manual errors, reduces administrative workload, and prevents proxy attendance. The system ensures real-time recording, secure data storage, and easy retrieval of attendance records. Its scalability and potential for future enhancements, such as cloud integration, biometric verification, and automated notifications, make it a reliable and modern approach to attendance management.

# REFERENCES

1. Arduino Official Website – Resources and documentation on Arduino Uno, sensors, and motor drivers
URL: https://www.arduino.cc/

2. Arduino IDE – Software for programming the Arduino URL: https://www.arduino.cc/en/software

3. Techatronic url link: https://techatronic.com/rfid-based-attendance-system-using-arduino/

4. ChatGPT

5. Prof. **Sahil Shukla**

# GLOSSARY

## Glossary:

- **RFID (Radio Frequency Identification):** A wireless technology that uses electromagnetic fields to automatically identify and track tags attached to objects or cards.
- **RFID Tag:** A small device or card containing a unique identification number that communicates with an RFID reader.
- **RFID Reader:** A device that scans RFID tags and sends the unique ID to a microcontroller or computer.
- **Arduino Uno:** A microcontroller board based on the ATmega328P used for interfacing sensors, readers, and other devices.
- **Microcontroller:** A compact integrated circuit that controls input/output devices and processes data in embedded systems.
- **Database:** A structured collection of data where attendance records are stored for retrieval and analysis.
- **Proxy Attendance:** When someone marks attendance on behalf of another person, often leading to inaccurate records.
- **PWM (Pulse Width Modulation):** A technique used in microcontrollers to control devices like motors or LEDs by varying signal width.
- **Real-Time System:** A system that processes and updates data immediately as events occur.